

ChE 382: Unit Operations Laboratory

Viscosity of a Non-Newtonian Fluid

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Introduction

Viscosity is a basic fluid property which is often required by engineers to make estimates of transport behavior such as mass transfer and heat transfer. It appears in many of the engineering correlations (e.g. Reynolds number). There are numerous compilations of viscosity values (see Reid, Sherwood, and Prausnitz [1]) for pure fluids and techniques to estimate viscosities for mixtures.

Theory [1], [2]

Viscosity is a fluid property and is defined as the ratio of the shearing stress on a fluid to the velocity gradient established in the fluid at the point of shear. The viscosity is a measure of the resistance of a fluid to a shearing force, or equivalently, the viscosity may be interpreted as a measure of the momentum flux in a fluid.

The physical meaning of viscosity can be easily understood by considering the flow between two parallel plates as illustrated in *Figure 1*. If one plate is moved at a constant velocity, a velocity profile is established in the fluid. It has been found experimentally that the force required to keep the plate in motion is proportional to the velocity gradient in the fluid. This relationship may be written as follows:

$$\tau_{yx} = -\mu \frac{dv_x}{dy} \quad (1)$$

Where τ_{yx} is the shear stress (i.e., force per unit area of fluid contacting the plate) in the +(positive)x direction on a surface with unit normal in the +y direction; dv_x/dy is the gradient in the y direction of the fluid velocity in the x direction; and μ is the fluid viscosity, which by definition is the ratio of the shear stress to the velocity gradient in the fluid normal to the shearing surface. The negative sign arises from convention; if the shear stress is taken to be positive (in the positive x direction), the velocity gradient established in the fluid is negative relative to the coordinate system used in *Figure 1*.

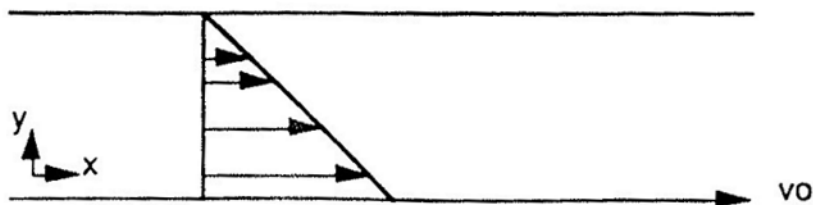


Figure 1. Velocity profile of a fluid contained between two parallel plates, one of which moving at a constant velocity v_0 .

Experimentally, however, it has been found that fluids can be categorized into a number of general classes. Newtonian fluids are those that obey *Equation (1)*. Non-Newtonian fluids, on the other hand, are fluids that do not exhibit a linear relationship between the shear stress and the velocity gradient. In these fluids, the apparent fluid viscosity will be a function of the rate of shear. The relationship between the shear stress and the velocity gradient shows a wide range of behavior for non-Newtonian fluids. Some examples are presented in *Figure 2*. Non-Newtonian fluids include: latex paints, polymer solutions, colloidal mixtures, foams, and two-phase fluids. Curves B and C in *Figure 2* are for fluids that are called pseudo-plastic and dilatant, respectively. For these fluids, a power-law relationship (also known as the Ostwald-deWaele Model) can adequately characterize the dependence of the shear rate on the velocity gradient:

$$\tau_{yx} = \pm m \left| \frac{dv_x}{dy} \right|^n \quad (2)$$

Here m and n are empirical constants and the $\pm$ sign is chosen such that the shear stress is in the correct direction relative to the velocity gradient. This equation is often written in the form:

$$\tau_{yx} = -m \left| \frac{dv_x}{dy} \right|^{n-1} \frac{dv_x}{dy} = -\mu_{app} \frac{dv_x}{dy} \quad (3)$$

Where μ_{app} is the apparent viscosity. Note that the apparent viscosity is not a constant but instead depends on the rate of shear experienced by the fluid. The actual relationship between the velocity gradient in the fluid and the shear stress is given in terms of the two constants, m and n defined by *Equation (2)*. For $n > 1$, the fluid behavior corresponds to curve C of *Figure 2*, whereas for $n < 1$, the fluid behavior is shown by curve B. For $n = 1$, *Equation (1)* is obtained with $m = \mu$. Curve A is for a Newtonian fluid and shows the linear relationship between velocity gradient and shear stress.

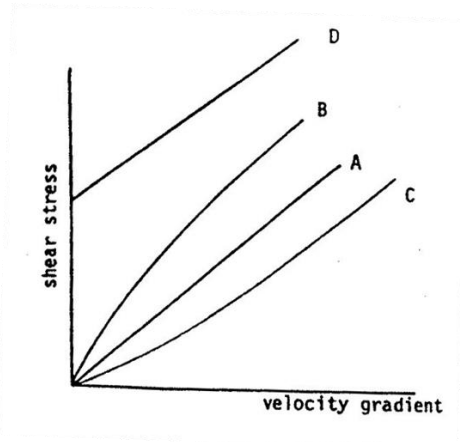


Figure 2. Relationship between strain and shear for different fluids: (A) Newtonian fluid; (B) psuedo-plastic fluid; (C) dilatant fluid; (D) Bingham fluid.

Experiment:

The Capillary Tube Viscometer

The flowrate and pressure drop across a capillary tube can be used to measure the viscosity of a fluid. The viscosity of a Newtonian fluid is related to the flowrate and pressure drop by the Hagen-Poiseuille equation:

$$Q = \frac{\Delta P \pi R^4}{8 \mu L} \quad (4)$$

Where Q is the volumetric flowrate of the fluid, ΔP is the pressure drop across the capillary, R and L the capillary radius and length, respectively, and μ the fluid viscosity. Thus measurements of Q vs. ΔP should yield a straight line of slope $\pi R^4 / 8 \mu L$, from which the viscosity may be calculated.

In this experiment, an analytical equation for the flowrate of a non-Newtonian fluid in a tube is used instead of Equation (4). For fluids that obey the Ostwald-deWaele viscosity model (Equation (3)), the following equation is obtained:

$$m \left[\frac{3n+1}{n} \right]^n \cdot \left(\frac{Q}{\pi R^3} \right)^n = \left(\frac{\Delta P}{2L} R \right) \quad (5)$$

Where m and n are the parameters of the Ostwald-deWaele model and the other terms are the same as in Equation (4). The derivation of the equation is given in appendix I.

A plot of $\log(\Delta P \cdot R / 2L)$ vs. $\log(Q / \pi R^3)$ will give a slope of ' n ' and an intercept value of $\log \left\{ m \left[\frac{3n+1}{n} \right]^n \right\}$. From the slope and intercept values obtained from the graph, the parameters m and n can then be calculated as a function of capillary diameter and length.

Experimental Procedure

The apparatus consists of a small capillary tube attached to a small pressurization vessel. The capillary tube is removable and the tubes of different length to diameter ratios may be attached to the vessel. The vessel is filled with the fluid for which the viscosity is to be measured and pressurized with air. Use caution when pressurizing and depressurizing the vessel. Always open the vent valve carefully. Open the vent valve first when depressurizing the vessel; keep the vent valve open while filling the reservoir. Always close the vent valve last when pressurizing the vessel. A constant pressure may be applied to the vessel by adjusting the pressure regulator on the air line.

Experiments will be conducted with CARBOPOL solution which is a non-Newtonian fluid. The CARBOPOL solution is prepared by mixing approximately two liters of a **1.0 wt% solution** of CARBOPOL in warm tap water. CARBOPOL 940 or 934 powder may be used.

Use the power mixer in the laboratory to dissolve the CARBOPOL powder thoroughly. Once the powder is completely dissolved in the water it will have a honey-like consistency. Add some 1N NaOH solution to the CARBOPOL solution to neutralize it to a pH of at least 5 to 7. The CARBOPOL will then become more gel-like. See supplemental information in Blackboard for more details on preparing Carbopol solutions and the viscosity range of CARBOPOL solutions as a function of pH.

Make experimental runs using several capillary tubes of different diameter and length. For each run, measure the flowrate at 3 to 4 different pressure differentials. The maximum pressure that should be used is 60 psig.

DATA ANALYSIS:

Explain how you neutralized the CARBOPOL solution and state the final pH value of the solution.

Make plots of $\log(\Delta P \cdot R/2L)$ vs. $\log(Q/\pi R^3)$ and determine values for 'm' and 'n' for the different capillary tubes that you used. What are the units of dimension for 'm' and 'n'? Do you obtain different results for capillaries having different L/D ratios? Why?

Determine what type of non-newtonian fluid the CARBOPOL solution is from your data.

Make a plot of the shear stress versus velocity gradient and compare to literature.

References

- [1] McCabe W.L., Smith J.C., Harriot P., *Unit Operations in Chemical Engineering*, 7th Edition pp. 46-52, (2005)
- [2] R.C. Reid, J.M. Prousnitz, T.K. Sherwood, *The Properties of Gases and Liquids*, 3rd ed. McGraw-Hill, 1977.
- [3] R.B. Bird, W.E. Stewart, and E.N. Lightfoot, *Transport Phenomena*, 2nd ed. John Wiley and Sons, Inc., 2002
- [4] J. Davidson, R. Clift, and D. Harrison, Eds., *Fluidization*, 2nd ed. Academic Press, 1995.
- [5] R.H. Perry and D.W. Green, Eds., *Perry's Chemical Engineer's Handbook*, 7th ed. McGraw-Hill, 1997.

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List of Symbols (SI units)

a = radius of the sphere (m)

F_b = buoyancy force (N)

F_D = drag force (N)

g = gravitational acceleration (m/s²)

K = calibration constant

L = length of capillary (m)

m = mass (kg)

Q = volumetric flowrate (m³/s)

R = radius of the capillary tube (m)

Re = Reynold's number (dimensionless)

t = time (s)

v = fluid velocity (m/s)

v_x = component of the fluid velocity in the x direction

ΔP = pressure drop (Pa)

μ = fluid viscosity (kg/m-s)

ρ_f = density of the fluid (kg/m³)

ρ_s = density of the sphere (kg/m³)

τ = shear stress (N/m²)

Appendix I.

From the equation of motion in cylindrical coordinates and the equation of continuity for an incompressible fluid one obtains:

$$\frac{dp}{dz} = -\frac{1}{r} \frac{d}{dr} (r \tau_{rz}) \quad (1)$$

for flow in a cylindrical tube. Here z is the direction of flow, r is the radial coordinate in the cylindrical tube, dp/dz is the pressure drop across the tube and τ_{rz} is the shear stress in the z direction on a fluid surface with unit normal in the r direction. dp/dz , the pressure drop across the tube, is a constant at steady flow. This term is equal to $-\Delta P/L$ where ΔP is the pressure drop for a tube of length L .

Integration of (1) with respect to z gives:

$$\tau_{rz} = \frac{\Delta p}{2L} r \quad (2)$$

Next, one substitutes the power law model for τ_{rz} to obtain:

$$\begin{aligned} \tau_{rz} &= -m \left| \frac{dv_z}{dr} \right|^{n-1} \frac{dv_z}{dr} \\ &= m \left| \frac{dv_z}{dr} \right|^n \end{aligned}$$

since dv_z/dr has a negative slope in the coordinate system for this problem. Therefore one obtains:

$$\frac{dv_z}{dr} = \left(\frac{1}{m} \frac{\Delta P}{2L} \right)^{1/n} r^{1/n} = ar^{1/n}$$

Separation of variables, integration, and application of the no-slip boundary condition that: at $r = R$, $v_z = 0$ gives:

$$\begin{aligned} \int_0^{v_z} dv_z &= a \int_R^r r^{1/n} dr \\ v_z &= a \frac{n}{n+1} r^{\frac{n+1}{n}} \Big|_R^r \\ v_z &= a \frac{n}{n+1} \left(r^{\frac{n+1}{n}} - R^{\frac{n+1}{n}} \right) \end{aligned}$$

The total flowrate, Q , is obtained by integration of the velocity profile over the tube cross section. In cylindrical coordinates this gives:

$$Q = \int_0^{2\pi} \int_0^R v_z r dr d\theta$$

$$\begin{aligned}
&= 2\pi a \left(\frac{n}{n+1} \right) \int_0^R \left(r^{\frac{n+1}{n}} - R^{\frac{n+1}{n}} \right) r dr \\
&= 2\pi a \left(\frac{n}{n+1} \right) \left[\frac{n}{3n+1} R^{\frac{n+1}{n}} - \frac{1}{2} R^{\frac{n+1}{n}} R^2 \right] \\
&= 2\pi a R^{\frac{3n+1}{n}} \left(\frac{n}{n+1} \right) \left[\frac{n}{3n+1} - \frac{1}{2} \right] \\
&= \pi a R^{\frac{3n+1}{n}} \left(\frac{n}{3n+1} \right)
\end{aligned}$$

$$\text{Or } Q = \pi \left[\frac{1}{m} \frac{\Delta P}{2L} \right]^{1/n} R^{\frac{3n+1}{n}} \left(\frac{n}{3n+1} \right)$$